

Point-Defect-Induced Half Metal in CrCl_3 Monolayer

Yuan Gao, Jing Wang,* Yan Li,* Meirong Xia, Zhiping Li, and Faming Gao

Electronic and magnetic properties of CrCl_3 monolayers are studied using density functional theory (DFT), including electron-electron Coulomb repulsion interactions (U). Our calculations indicate CrCl_3 single layer is achievable from easily cleavable, layered and insulating CrCl_3 bulk. The CrCl_3 monolayer is ferromagnetic and the magnetic order would be enhanced by a Cr vacancy. Most importantly, the point defect will introduce an insulator-to-half-metal transition, making it a promising candidate for spintronics in room temperature. Atomic point defects are omnipresent in materials and this study paves a new way to realize 100% spin polarization.

Introduction: Transition metal compounds are of great interests for decades and many spectacular phenomena have been discovered in this family, that is, superconductors,^[1] multi-ferroics,^[2] and magnetoresistance.^[3] In recent years, much attention has been concentrated on spintronics and a number of transition metal compounds are predicted to be half metallic. Half metal (HM) was first defined by de Groot et al. and was found to be asymmetrical in band structure,^[4] that is, being metallic in one spin channel while insulating (semiconducting) in the other. Half metals can generate 100% spin polarized currents without any external operation and play significant roles in the discipline of spintronics.^[5] Most transition metals are intrinsic magnetic which tend to play distinct behaviors in different channels and this contributes to the half metal. Moreover, some transition metal compounds,^[6,7] especially those containing 3d electrons, have high magnetic temperature and they are promising candidates for room temperature technologies.

On the other hand, substantial efforts have been made to integrate novel properties of low dimensional materials since the synthesis of graphene.^[8–10] Thanks to the technologies of deposition and chemical exfoliation, a wide range of two-dimensional(2D) compounds have been manufactured and reported to behave diversely compared with their bulk counterparts. The canonical example is the graphite-graphene case, but monolayers have been exfoliated from many other compounds as well,^[11–13] such as monolayers of hexagonal boron nitride,^[14] transition metal chalcogenides MoS_2 and WS_2 (ref. [15,16]) and phosphorene.^[17] Away from serving as a platform for

manipulating functionalities and properties, 2D materials meet the requirement accelerating size reduction and efficiency improvements in microelectronics.

Under above considerations, efforts on 2D HFs are of considerable importance. Furthermore, although HFs displace superior properties and meet the “basic” demand of spintronics applications, some issues, such as stability in room temperature and long distance spin transport, still remain to exploit the full potential of spintronics in information transfer and storage. In an effort to discover novel 2D

HMs, transition metal chlorides is studied in this paper. Bulk CrCl_3 is a semiconductor.^[18] And so does CrCl_3 monolayer from our calculations under the framework of density functional theory (DFT). To achieve half metallic quantum state, a point defect with Cr vacancy is simulated. In fact, an insulator-(semiconductor) to HM transition by band engineering is a new route to realize 100% spin polarization.^[19–22] To this end, several methods have been carried out. For example, monolayer VS_2 experiences an electronic phase transition from a bipolar magnetic semiconductor (BMS) to a half metal via hole doping.^[19] Wu et al. reported an interesting transition from a high spin (HS) antiferromagnetic insulator to an intermediate-spin (IS) ferromagnetic HM in layered cobaltate $\text{Sr}_2\text{CoO}_3\text{F}$, by a pressure of 5.4 GPa.^[20] In previous studies of the authors,^[21,22] uniaxial pressure and antisite defects are also found to be useful to turn off the small band gap in one spin channel meanwhile maintaining the large one in the other, realizing half metallic character.

Computational Methods: Structure optimization and electronic structure calculations were done by VASP,^[23–26] based on the projector augmented-wave (PAW) method.^[27,28] For the self-consistent calculation with the PAW, the plane wave cut-off energy was chosen to be 500 eV. For bulk(monolayer) CrCl_3 , a mesh of $3 \times 3 \times 3$ ($3 \times 3 \times 1$) *k*-points were used for geometry optimization and a finer one with $5 \times 5 \times 3$ ($5 \times 5 \times 1$) *k*-points were adopted for calculating electronic and magnetic properties. The convergence of these *K*-points was tested to make sure the calculation is unambiguous. For single layer of CrCl_3 , a vacuum space of 15 Å is employed and Van der Waals interaction (vdW) correlation is considered by using the semiempirical dispersion-corrected density functional theory (DFT-D2) force-field approach.^[29,30] For the self-consistent field iterations, the convergence tolerances were selected as the difference in total energy and the maximum force being within 1.0×10^{-5} eV and 1.0×10^{-2} eV Å⁻¹, respectively. The exchange-correlation energy was treated by the generalized gradient approximation (GGA-PBE).^[31] Electron-electron Coulomb repulsion interactions (U) for transition metals in combination with the double-counting correction were

Y. Gao, Prof. J. Wang, Dr. Y. Li, Dr. M. Xia, Prof. Z. Li, Prof. F. Gao
Key Laboratory of Applied Chemistry
Yanshan University
Qinhuangdao 066004, P. R. China
E-mail: jwang6027@ysu.edu.cn; liyan2522@ysu.edu.cn

The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/pssr.201800105>.

DOI: 10.1002/pssr.201800105

considered in the rotationally invariant form (GGA+U) with a single effective Hubbard parameter $U_{\text{eff}} = U - J$.^[32] A series of U values were selected, that is, 3.0–6.0 eV for Cr. Our calculations indicated that for the selected U values, the main conclusions are consistent. Therefore, we choose the results from $U_{\text{Cr}} = 5.0$ eV for discussion which is a typical value for 3d electrons.^[21,22,33] The results from other U values are provided in the supporting (Table S1 and Figure S1 and S2).

Results and Discussion: Chromium trihalides CrX_3 ($X = \text{Cr}, \text{Br}, \text{I}$), like MoS_2 , are another class of van der Waals (vdW) bonded, layered compounds,^[34–36] providing possibility to produce monolayers by exfoliation. Specifically, this paper focuses on CrCl_3 which crystallizes in monocline AlCl_3 structure (space group: $C2/m$) at room temperature and rhombohedral BiI_3 type (space group: $R-3$) below 240 K^[34–36] (Figure 1a and b). Both symmetries were examined including different magnetic coupling manners, that is, ferromagnetic (FM, the spins of all Cr parallelized), antiferromagnetic (AFM, the spins of neighboring Cr antiparallelized) and layered antiferromagnetic (L-AFM, the spins of Cr in the same layer parallelized while they are antiparallelized between layers). As seen in Figure 2 in which the energy of rhombohedral $R-3$ with FM is set to be zero, $C2/m$ with L-AFM is favored in GGA method as it is below zero. While including U , rhombohedral $R-3$ with FM becomes the most stable state regardless the value of U . However, it is noticed that the energy differences from the L-AFM in both $R-3$ and $C2/m$ symmetries are minor (only several meV), which underlines all these phases might coexist and this can explain why different experiments argue different crystal

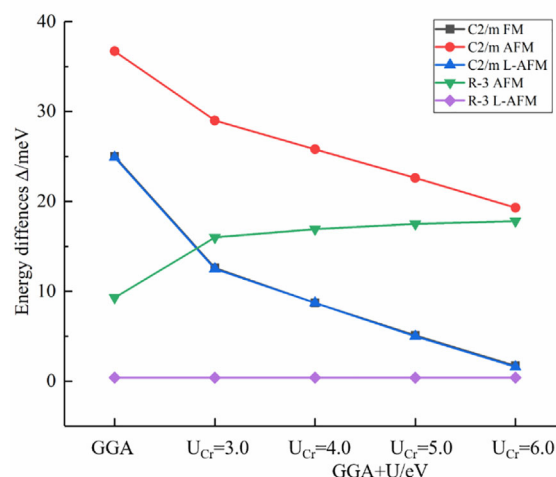


Figure 2. Energy differences (meV) from different symmetries with different magnetic coupling for bulk CrCl_3 from GGA and GGA+U methods.

structures.^[34–36] In fact, Cr^{3+} ions are arranged in a honeycomb network in edge-sharing octahedral coordination by six Cl^- ions in layers in both monoclinic and stacking sequence along z direction. As a result, after exfoliating, only one type of monolayer can be achieved (Figure 1c). The calculated structure parameters as well as those existing in literatures are summarized in Table 1, which are consistent with each others for bulk phases.

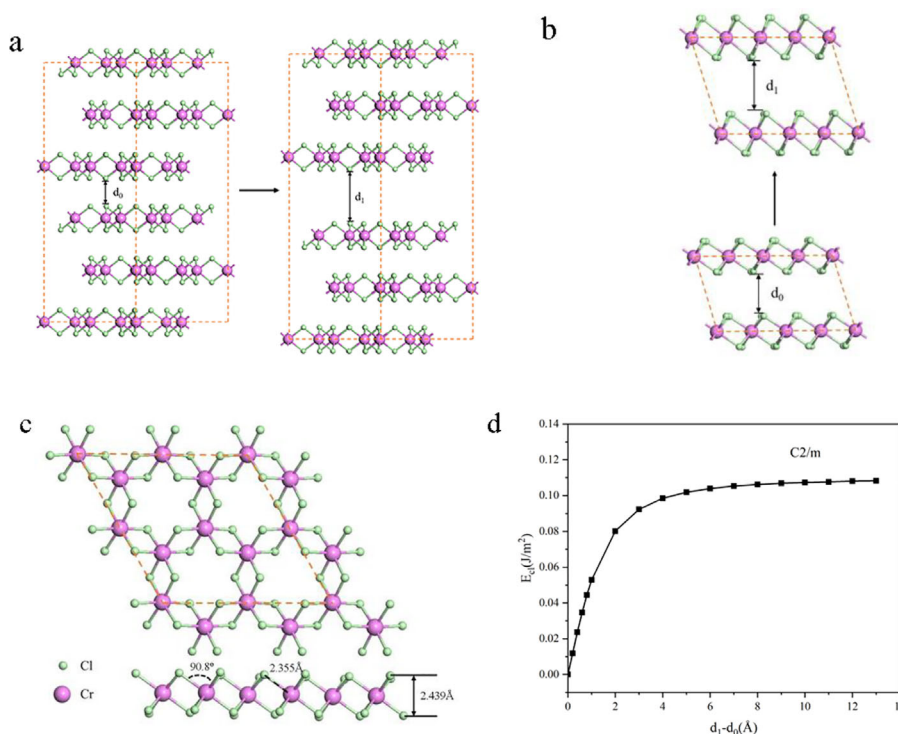


Figure 1. Side view of bulk CrCl_3 with rhombohedral (a) and monoclinic structure (b) and corresponding fracture used to simulate the exfoliation procedure; (c) top and side views of the CrCl_3 monolayer; (d) cleavage energy E_{cl} as a function of the separation between two fractured parts.

Table 1. Structure parameters for bulk (*C2/m* and *R-3*) and CrCl_3 monolayer (*P-31m*) from our calculations and literatures.

	Our calculations			a)		b)
Space group	<i>C2/m</i>	<i>R-3</i>	<i>P-31M</i>	<i>C2/m</i>	<i>R-3</i>	<i>P-31M</i>
<i>a</i> (Å)	5.985	5.980	5.985	5.959	5.942	6.108
<i>b</i> (Å)	10.357	5.980	5.985	10.321	5.942	6.108
<i>c</i> (Å)	6.200	17.595	2.439	6.114	17.333	2.698
(°)	108.1	90.0	90.0	108.5	90.0	90.0
Cr-Cl (Å)	2.353×2	2.355×3	2.355×6	2.328×3	2.370×4	2.380×6
	2.354×2	2.354×3		2.364×3	2.371×2	
	2.356×2					

a) ref. [35]; b) ref. [35].

Table 2. Magnetic moment (μ_B) and energy differences ($\Delta E = E_{\text{FM}} - E_{\text{AFM(FiM)}}$, meV) for the perfect and defect monolayer $\text{Cr}_7\text{Cl}_{24}$ from GGA and GGA + U ($U_{\text{Cr}} = 5.0$ eV) methods.

		GGA	GGA + U
CrCl_3	Cr	2.86	3.15
	Cl	−0.03	−0.07
	ΔE	−7.0	−15.4
CrCl_3 with a Cr vacancy	Cr_1	2.83	3.14
	Cr_2	2.27	3.02
	Cl_1	−0.03	−0.08
	Cl_2	−0.09	−0.29
	ΔE	−183.6	−198.9

Regarding the CrCl_3 monolayer, the in-plane parameter *a* and *b* are almost same as in bulk phase from our calculations. The bond lengths, although have small differences in bulk, are all 2.355 Å which would give a perfect octahedron in the monolayer. The *c* (2.439 Å) means the thickness of the layer.

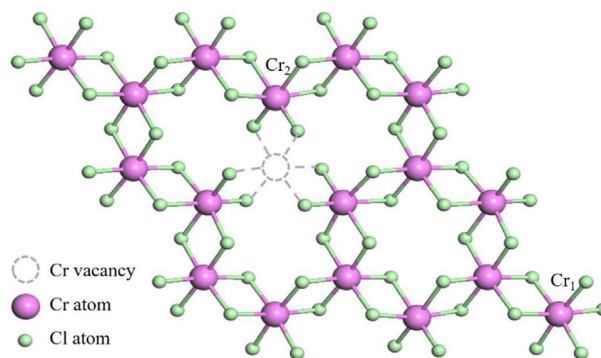


Figure 4. Top views of the geometric structure of $\text{Cr}_7\text{Cl}_{24}$ monolayer with a Cr vacancy.

To check whether it is feasible to exfoliating a monolayer from bulk CrCl_3 , cleavage energy is calculated by introducing a fracture in the bulk (Figure 1d). Total energy is measured and it is found to be increased gradually at first with the increasing separation between the two fractured parts. It then converges slowly at the ideal cleavage cohesion energy of about 0.11 J m^{-2} , close to previous study.^[37] This value is smaller than that of graphite ($\approx 0.36 \text{ J m}^{-2}$), indicating the feasibility of exfoliation from bulk CrCl_3 . This is verified by a recent experiment in which monolayer and few layer CrCl_3 is successfully prepared.^[38]

Spin-polarized calculations are performed to check whether the FM state survives in the CrCl_3 monolayer. Both FM and AFM coupling are tested and it is found that FM is lower in energy (Table 2). The spin magnetic moment for Cr^{3+} is $2.86 \mu_B$ in GGA calculation and it increases to be $3.15 \mu_B$ in GGA+U ($U = 5.0$ eV), close to its theoretical value ($3d^3$, $S = 3/2$) and previous studies.^[35,37] The total magnetic moment per unit cell is $3.0 \mu_B$, underlying a semiconductor or a HM. Besides the magnetic coupling manner, semiconducting nature is also reserved in the single layer (Figure 3). The band gap in the vicinity of the Fermi level (E_F) is ≈ 1.6 eV in GGA calculation and a close inspection find that the gap is formed between $\text{Cr } t_{2g}$ spin up state and e_g

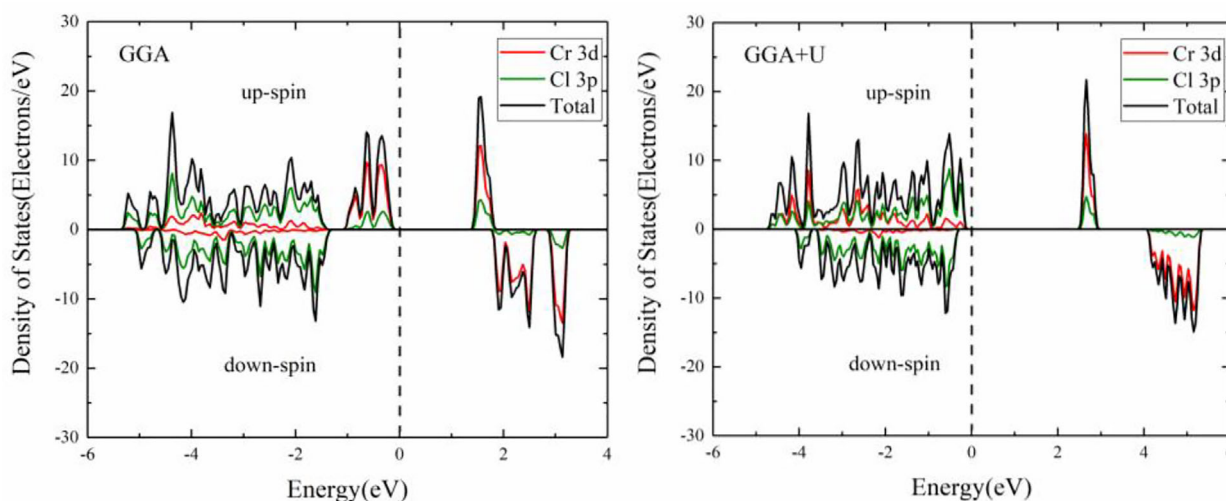


Figure 3. Density of states for CrCl_3 monolayer from the GGA and GGA+U ($U_{\text{Cr}} = 5.0$ eV) methods.

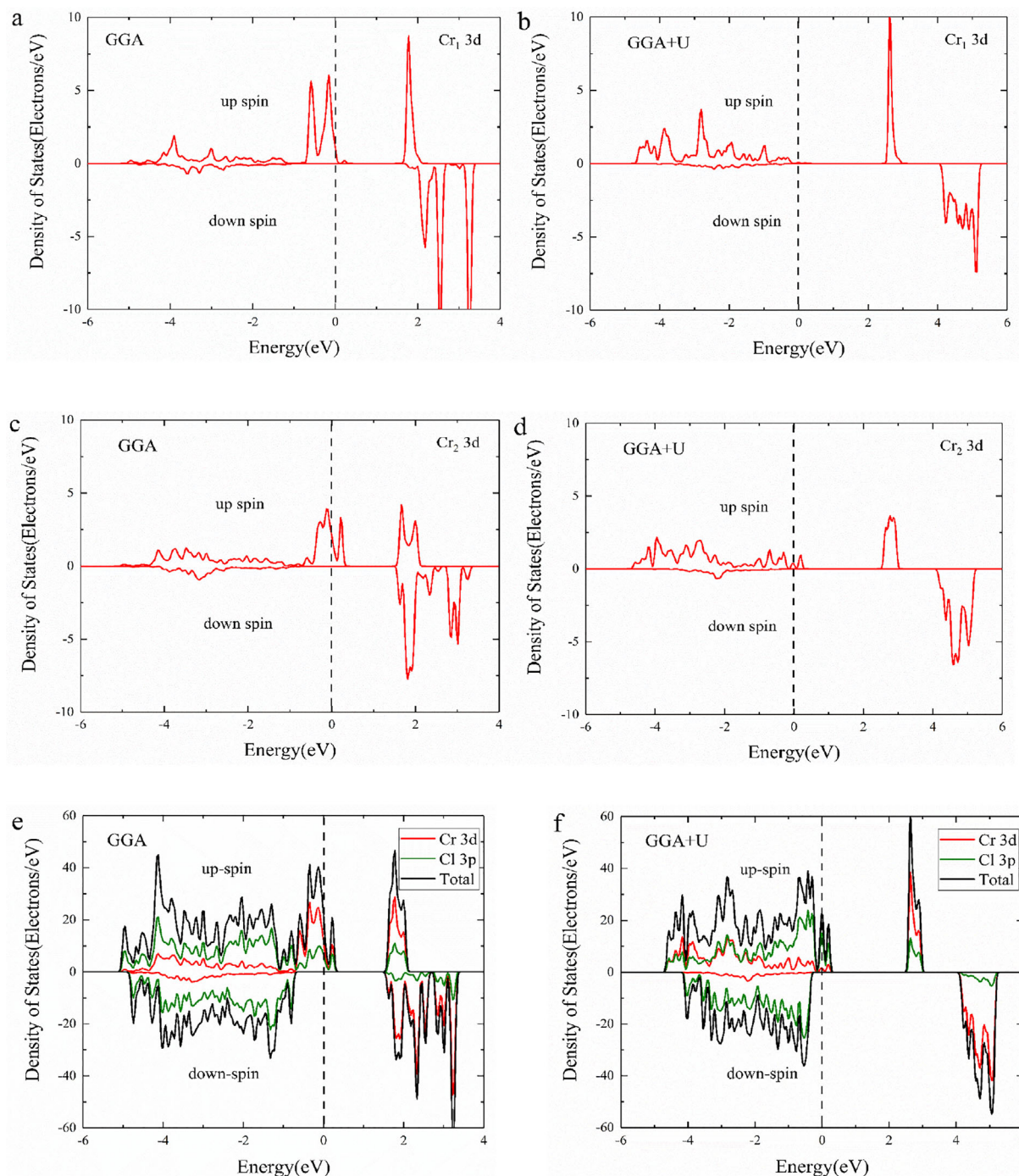


Figure 5. Density of states for $\text{Cr}_7\text{Cl}_{24}$ monolayer from the GGA and GGA+U methods.

spin up state. It is enlarged to be 2.2 eV in GGA+U ($U = 5.0$ eV) method because the occupied Cr t_{2g} state is pushed downwards and the unoccupied states moving to opposite direction. As a result, the gap is between Cl 3p spin state and Cr 3d unoccupied state.

In recent years, band engineering is introduced to achieve HM by the means of loading pressure,^[39] uniaxial strain,^[40] electron/hole doping,^[41] defects,^[42] etc. Typically, atomic point defects are omnipresent in materials and physical properties of

functional materials are strongly influenced. Therefore, to achieve HM for monolayer CrCr_3 , a Cr vacancy is simulated by taking away one Cr ion from a $\text{Cr}_8\text{Cl}_{24}$ fragment. The structure of the resultant $\text{Cr}_7\text{Cl}_{24}$ is optimized and the lattice parameters do not change much ($a = b = 5.949$ Å), but the local structure close to the Cr defect is different. Taking Cr_2 (Figure 4) for an example which is the nearest neighbor of the vacancy, the bond lengths can be divided into three groups and they are 2.176, 2.446, and 2.352 Å, respectively. For those (such as Cr_1) far away from the

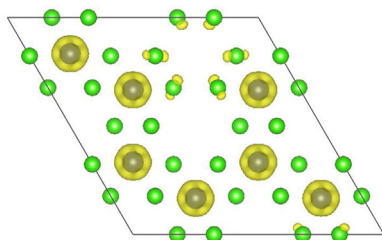


Figure 6. Charge density for $\text{Cr}_7\text{Cl}_{24}$ monolayer from GGA+U method.

vacancy they remain almost unchanged. Thus, it is not unexpected that the electron state for those Cr ions in the vicinity of the vacancy will behave quite different as a result of the local chemical environment change. As shown in **Figure 5a–d**, a few electrons of Cr_2 leak out of the E_F thus the spin up channel is metallic while the electron state of Cr_1 is similar as in perfect case. This is understandable in view of the fact that the bond distance is compressed in Cr_2 which is unable to stand as much electron as before. On the other hand, the valence of the cationic ions has to increase to balance the valence of anion after one Cr missing. To do that, the amount of electron would be reduced in certain ions, and these refer to Cr_2 whose local chemical environment changes significantly. As a consequence, in contrast to the pristine monolayer, the case with a Cr vacancy shows HM nature in both GGA and GGA+U methods- (Figure 5e,d). Local structure change induced electronic variation is important in discovering half metal and similar transition is found in other cases such as BaCl .^[43,44]

Turning to the magnetic properties, the magnitude of spin moment for Cr_2 is smaller than Cr_1 . that is, it is $2.27 \mu_B$ for the former while it is $2.83 \mu_B$ for the latter in GGA method (Table 2). With electron correlation (U) included, the difference become smaller. On the other hand, distinct magnetic moment is observed at Cl sites which are close to the defects and the magnitude is as large as $0.29 \mu_B$ (GGA + U). In fact, finite spin density difference is found on these Cl atoms (**Figure 6**). Induced magnetic moment on nonmagnetic ions is found in many cases. In $\text{Sr}_2\text{FeOsO}_6$, for instance, small magnetic moment is induced on oxygen ions.^[45] The magnetic coupling of the whole system would be enhanced by these induced magnetic moments.

It is noticed that although the energy difference between FM and AFM state is small in the perfect CrCl_3 monolayer (7.0 meV in GGA and 15.4 meV in GGA+U, Table 2), it is enlarged in the defect case (183.6 meV in GGA and 198.9 meV in GGA+U, Table 2). According to the mean field theory, the magnetic ordering temperature (T) is scaled to the energy difference ($T = \Delta E / 3k_B$), thus the Néel temperature (T_N) is estimated to be 710 K in GGA and 769 K in GGA+U method, much higher than room temperature. This means the defect CrCl_3 monolayer is a promising HM in real. The enhanced magnetic ordering temperature is understandable in view of the fact that the FM coupling is strengthened by the induced magnetic moment on Cl ions.

Finally, to investigate whether the half metal nature would survive for lower defect concentration, two larger cases, i.e., $\text{Cr}_{17}\text{Cl}_{54}$ and $\text{Cr}_{31}\text{Cl}_{96}$ are simulated. The defect concentration for the former is 5.56% while that for the latter is 3.12%, which are more feasible for experiment. It is found that similar

conclusions can be made for both cases (Supporting Information, Figure S3) and this indicates the half metallic character is independent of the defect concentration. This is understandable that the influence on the electronic structure is local, as in the study of MnO_2 .^[46] However, in view of the lack of the exact concentration of defect concentration from experiments, the study deserves further investigation.

Conclusions: In summary, half metallic quantum state is predicted in the CrCl_3 monolayer with a Cr point defect from our first-principles calculations. The half metallic state is stable regardless of U values and considered defect concentration, that is, from 3.12 to 12.5%. Moreover, induced magnetic moment is found at Cl ion sites and it will strengthen the ferromagnetic coupling between Cr ions, making it a promising candidate for room temperature spintronics.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

The authors thank the NSFC (Nos. 21403185), NSF of Hebei (No. B2015203268), and Yanshan University (No. 13LGB026, 14LGA017) for financial support.

Conflict of Interest

The authors declare no conflict of interest.

Keywords

2D materials, density functional theory, ferromagnetism, half metals

Received: March 7, 2018

Revised: April 11, 2018

Published online:

- [1] W. E. Pickett, *Phys. Rev. Lett.* **1996**, 77, 3185.
- [2] R. Ramesh, N. A. Spaldin, *Nat. Mater.* **2007**, 6, 21.
- [3] S. Ikeda, J. Hayakawa, Y. Ashizawa, Y. M. Lee, K. Miura, H. Hasegawa, M. Tsunoda, F. Matsukura, H. Phno, *Appl. Phys. Lett.* **2008**, 93, 082508.
- [4] R. A. de Groot, F. M. Mueller, P. G. van Engen, K. H. J. Buschow, *Phys. Rev. Lett.* **1983**, 50, 2024.
- [5] S. A. Wolf, D. D. Awschalom, R. A. Buhrman, J. M. Daughton, S. von Molnar, M. L. Roukes, A. Y. Chtchelkanova, D. M. Treger, *Science* **2001**, 294, 1488.
- [6] N. S. Rogado, J. Li, A. W. Sleight, M. A. Subramanian, *Adv. Mater.* **2005**, 17, 2225.
- [7] R. N. Mahato, K. Sethupathi, V. Sankaranarayanan, *J. Appl. Phys.* **2010**, 107, 09D714.
- [8] F. Mhoeini, Kh. Shakouri, F. M. Peeters, *Phys. Rev. B* **2016**, 94, 125412.
- [9] K. S. Novoselov, A. K. Geim, S. V. Morozov, D. Jiang, M. I. Katsnelson, I. V. Grigorieva, S. V. Dubonos, A. A. Firsov, *Nature (London)* **2005**, 438, 197.

- [10] Y. Zheng, Y. W. Tan, H. L. Stormer, P. Kim, *Nature (London)* **2005**, 438, 201.
- [11] M. Ashton, J. Paul, S. B. Sinnott, R. G. Hennig, *Phys. Rev. Lett.* **2017**, 118, 106101.
- [12] P. Joensen, R. Frindt, S. R. Morrison, *Mater. Res. Bull.* **1986**, 21, 457.
- [13] Y. Lin, T. V. Williams, J. W. Connell, *J. Phys. Chem. Lett.* **2010**, 1, 277.
- [14] J. N. Coleman, M. Lotya, A. ÓNeill, S. D. Bergin, P. J. King, U. Khan, K. Young, A. Gaucher, S. De, R. J. Smith, *Science* **2011**, 331, 568.
- [15] K. F. Mak, C. Lee, J. Hone, J. Shan, T. F. Heinz, *Phys. Rev. Lett.* **2010**, 105, 136805.
- [16] Q. H. Wang, K. Kalantar-Zadeh, A. Kis, J. N. Coleman, M. S. Strano, *Nat. Nanotechnol.* **2012**, 7, 699.
- [17] H. Liu, A. T. Neal, Z. Zhu, Z. Luo, X. F. Xu, D. Románek, P. D. Ye, *ACS Nano* **2014**, 8, 4033.
- [18] C. Starr, F. Bitter, A. R. Kaufmann, *Phys. Rev.* **1940**, 58, 977.
- [19] N. N. Luo, C. Si, W. H. Duan, *Phys. Rev. B* **2017**, 95, 205432.
- [20] X. D. Ou, F. R. Fan, Z. W. Li, H. B. Wang, H. Wu, *Appl. Phys. Lett.* **2016**, 108, 092402.
- [21] J. Wang, X. F. Hao, Y. H. Xu, Z. P. Li, N. N. Zu, Z. J. Wu, F. M. Gao, *RSC Adv.* **2015**, 5, 50913.
- [22] J. Wang, L. J. Zhai, Y. Gao, Z. P. Li, Y. H. Xu, X. F. Hao, F. M. Gao, *Phys. Status Solidi RRL* **2017**, 11, 1700052.
- [23] G. Kresse, J. Hafner, *Phys. Rev. B* **1993**, 47, 558.
- [24] G. Kresse, J. Hafner, *Phys. Rev. B* **1994**, 49, 14251.
- [25] G. Kresse, J. Furthmüller, *Comput. Mater. Sci.* **1996**, 6, 15.
- [26] G. Kresse, J. Furthmüller, *Phys. Rev. B* **1996**, 54, 11169.
- [27] P. E. Blöchl, *Phys. Rev. B* **1994**, 50, 17953.
- [28] G. Kresse, D. Joubert, *Phys. Rev. B* **1999**, 59, 1758.
- [29] S. Grimme, *J. Comput. Chem.* **2006**, 27, 1787.
- [30] S. Grimme, J. Antony, S. Ehrlich, H. Krieg, *J. Chem. Phys.* **2010**, 132, 154104.
- [31] J. P. Perdew, K. Burke, M. Ernzerhof, *Phys. Rev. Lett.* **1996**, 77, 3865.
- [32] A. I. Liechtenstein, V. I. Anisimov, J. Zaanen, *Phys. Rev. B* **1995**, 52, R5467.
- [33] H.-S. Jin, K.-W. Lee, *Phys. Rev. B* **2011**, 84, 172405.
- [34] C. N. Avram, A. S. Gruia, M. G. Brik, A. M. Barb, *Phys. B* **2015**, 478, 31.
- [35] H. Wang, V. Eyert, U. Schwingenschlogl, *J. Phys.: Condens. Matter* **2011**, 23, 116003.
- [36] B. Morosin, A. Narath, *J. Chem. Phys.* **1964**, 40, 1958.
- [37] A. M. McGuire, H. Dixit, V. R. Cooper, B. C. Sales, *Chem. Mater.* **2015**, 27, 612.
- [38] M. A. McGuire, G. Clark, K. C. Santosh, W. M. Chance, G. E. Jellison, V. R. Cooper, X. D. Xu, B. C. Sales, *Phys. Rev. Mater.* **2017**, 1, 014001.
- [39] K.-W. Lee, W. E. Pickett, *Phys. Rev. B* **2011**, 83, 180406(R).
- [40] G. Sclausero, K. Dymkowski, C. Ederer, *Phys. Rev. B* **2016**, 94, 245109.
- [41] A. Garg, H. R. Krishnamurthy, M. Randeria, *Phys. Rev. Lett.* **2014**, 112, 106406.
- [42] M. D. Mukadam, S. Roy, S. S. Meena, P. Bhatt, S. M. Yusuf, *Phys. Rev. B* **2016**, 94, 214423.
- [43] M. J. Greschner, D. D. Klug, Y. S. Yao, *Phys. Rev. B* **2016**, 93, 094428.
- [44] Y. S. Yao, D. D. Klug, *Phys. Rev. B* **2013**, 88, 014113.
- [45] J. Wang, N. N. Zu, X. F. Hao, Y. H. Xu, Z. P. Li, Z. J. Wu, F. M. Gao, *Phys. Status Solidi RRL* **2014**, 8, 776.
- [46] H. Wang, J. J. Zhang, X. D. Hang, X. D. Zhang, J. F. Xie, B. C. Pan, Y. Xie, *Angew. Chem.* **2014**, 126, 1.